

Group: Let G be a non-empty set and let $\star$ be a binary operation on G . The algebraic object $(G, \star)$ is called a group if it satisfies the following properties:

i) $\star$ is associative on G . i.e.

$$a \star (b \star c) = (a \star b) \star c \quad \forall a, b, c \in G.$$

ii) G has an identity. i.e.

$\exists$ an element $e \in G$ such that

$$a \star e = a = e \star a \quad \forall a \in G.$$

iii) for each $a \in G$, $\exists$ an element $b \in G$ such that

$$a \star b = e = b \star a.$$

i.e. a has an inverse.

Remark: We know ~~that~~ there exists unique identity and inverse.

Example: $(\mathbb{Z}, +)$ is a group.

i) $+$ is associative on $\mathbb{Z}$.

ii) 0 is the identity of $(\mathbb{Z}, +)$

iii) $-a \in \mathbb{Z}$ is the inverse of $a \in \mathbb{Z}$ under $+$.

Defⁿ: A group $(G, \star)$ is called commutative (or abelian) if

$$a \star b = b \star a \quad \forall a, b \in G.$$

A group $(G, *)$ is called a finite group if the cardinality $|G|$ is finite, otherwise it is called an infinite group.

$|G|$ is called the order of the group $(G, *)$.

Example: 1) $(\mathbb{Z}, +)$ is an infinite abelian group.

2) $(\mathbb{Z}, \cdot)$ is not a group.

i) $\cdot$ is associative on $\mathbb{Z}$.

(ii) 1 is the identity of $(\mathbb{Z}, \cdot)$.

(iii) Let $a \in \mathbb{Z}$. & let $b \in \mathbb{Z}$ be the inverse of a in $(\mathbb{Z}, \cdot)$

Then $a \cdot b = 1 \Rightarrow a = b = 1$ or $a = b = -1$.

$\therefore$ only 1 and -1 has inverse in $(\mathbb{Z}, \cdot)$.

Hence $(\mathbb{Z}, \cdot)$ is not a group.

3) $G = \{1, -1, i, -i\}$ is an abelian group of order 4 under the usual multiplication of complex numbers.

i) $\cdot$ is associative on G .

ii) 1 is the identity

iii) 1 has inverse 1 , -1 is inverse of -1 .

i & $-i$ are inverses of each other

Example: Cayley table of $G = \{1, -1, i, -i\}$ under usual multiplication $\cdot$ of complex numbers.

	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

this table shows that G is closed under $\cdot$
i.e. $\cdot$ is a binary operation on G .

The subset $H = \{1, -1\} \subseteq G$ is closed under $\cdot$.

This can be easily checked from the above table.

① Let $\mathbb{Q}^+$ be the set of positive rational numbers and let $\cdot$ be the multiplication operation. $(\mathbb{Q}^+, \cdot)$ is a group.

- i) $\cdot$ is associative on $\mathbb{Q}^+$.
- ii) 1 is the identity
- iii) for each $a \in \mathbb{Q}^+$, $\frac{1}{a}$ is the inverse of a .

Let $\star$ be a binary operation on a non-empty set G .
A subset $H (\subseteq G)$ is ~~also~~ called closed under $\star$ if

$$a \star b \in H \quad \forall a, b \in H.$$

Eg: '+' is a binary operation on $\mathbb{Q}$, the set of all rational numbers.

$\mathbb{Z} \subseteq \mathbb{Q}$ is closed under '+',

$\mathbb{N} \subseteq \mathbb{Q}$ is closed under '+',

Remark: $\star$ is a binary operation on G is equivalent of saying G is closed under $\star$.

Let G be a ~~set~~ non-empty set, and $\star$ be an operation defined on G .

Question: How to check whether $\star$ is a binary operation G or not? or in other words is G closed under $\star$ or not.

Cayley table (or multiplication table) helps to decide that.

Definition: Let $G = \{g_1, g_2, \dots, g_n, \dots\}$ be a set. Let $\star$ is defined on $G \times G$. The Cayley table of G is the matrix whose (i, j) th entry is the element $g_i \star g_j$.

⑤ Let G be the set of all positive irrational numbers and let $\cdot$ is the usual multiplication operation. Then $(G, \cdot)$ is not a group.

G is not closed under $\cdot$ or $\cdot$ is not a binary operation on G as

$$\sqrt{2} \in G \quad \& \quad \sqrt{2} \cdot \sqrt{2} = 2 \notin G.$$

⑥ Let $M_n(\mathbb{R})$ be the set of all $n \times n$ matrix over $\mathbb{R}$.
Let $+$ be the matrix addition operation on $M_n(\mathbb{R})$.

i) $+$ is associative on $M_n(\mathbb{R})$ as

$$A + (B + C) = (A + B) + C \quad \forall A, B, C \in M_n(\mathbb{R}).$$

ii) $O = \begin{bmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix}_{n \times n} \in M_n(\mathbb{R})$ is the identity.

iii) Let $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \in M_n(\mathbb{R}).$

$-A = \begin{bmatrix} -a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & -a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \dots & -a_{nn} \end{bmatrix}$ is the inverse of A under $+$.

$(M_n(\mathbb{R}), +)$ is a group.

⑦ Let us consider the matrix multiplication ' $\cdot$ ' on the set $M_n(\mathbb{R})$.

i) ' $\cdot$ ' is associative on $M_n(\mathbb{R})$ as

$$A \cdot (B \cdot C) = (A \cdot B) \cdot C \quad \forall A, B, C \in M_n(\mathbb{R}).$$

ii) $I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} \in M_n(\mathbb{R})$ is the identity.

iii) $A \in M_n(\mathbb{R})$ has an inverse if and only if $\det(A) \neq 0$.

So $(M_n(\mathbb{R}), \cdot)$ is not a group.

Consider the set $GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$.

Consider $(GL(n, \mathbb{R}), \cdot)$.

' $\cdot$ ' is a binary operation on $GL(n, \mathbb{R})$ i.e.,

$GL(n, \mathbb{R})$ is closed under ' $\cdot$ ' as,

$$A, B \in GL(n, \mathbb{R}) \Rightarrow \det(A) \neq 0 \quad \det(B) \neq 0$$

$$\det(A \cdot B) = \det(A) \cdot \det(B) \neq 0.$$

So $A \cdot B \in GL(n, \mathbb{R})$.

Hence $(GL(n, \mathbb{R}), \cdot)$ is a group. It is called the general linear group of all $n \times n$ matrices over $\mathbb{R}$.

$$\textcircled{8}. \text{ Let } SL(n, \mathbb{R}) = \{ A \in M_n(\mathbb{R}) \mid \det(A) = 1 \}$$

Consider $(SL(n, \mathbb{R}), \cdot)$ where $\cdot$ is the matrix multiplication.

$$\text{Let } A, B \in SL(n, \mathbb{R}). \Rightarrow \det(A) = 1 = \det(B).$$

$$\text{So } \det(A \cdot B) = \det(A) \cdot \det(B) = 1 \cdot 1 = 1.$$

$$\Rightarrow A \cdot B \in SL(n, \mathbb{R}).$$

$\Rightarrow \cdot$ is a binary operation on $SL(n, \mathbb{R})$ or $SL(n, \mathbb{R})$ is closed under $\cdot$.

i) $\cdot$ is associative.

ii) I is the identity.

$$\text{iii) Let } A \in SL(n, \mathbb{R}). \Rightarrow \det(A) = 1 \neq 0.$$

So A^{-1} exists.

$$A \cdot A^{-1} = I.$$

$$\Rightarrow \det(A \cdot A^{-1}) = \det(I)$$

$$\Rightarrow \det(A) \cdot \det(A^{-1}) = 1 \Rightarrow \det(A^{-1}) = 1.$$

$$\text{as } \det(A) = 1 = \det(I).$$

$$\Rightarrow A^{-1} \in SL(n, \mathbb{R}).$$

So every element $A \in SL(n, \mathbb{R})$ has an inverse in $SL(n, \mathbb{R})$. Hence $(SL(n, \mathbb{R}), \cdot)$ is a group.

⑨ $(\mathbb{Z}, -)$ is not a group as $'-'$ is not associative on $\mathbb{Z}$.

⑩. Let $\mathbb{R}^* = \mathbb{R} \setminus \{0\}$ be the set of all non-zero real numbers.

Consider $(\mathbb{R}^*, \cdot)$.

i) $'\cdot'$ is associative on $\mathbb{R}^*$.

ii) 1 is the identity

iii) $a \in \mathbb{R}$ has the inverse $\frac{1}{a} \in \mathbb{R}$.

So $(\mathbb{R}^*, \cdot)$ is a group.

⑪ Let X be a non-empty set & let $S(X)$ be the set of all bijections from X onto itself.

$$\text{i.e. } S(X) = \{f: X \rightarrow X \mid f \text{ is a bijection}\}.$$

Consider the composition of functions operation $'\circ'$ on $S(X)$, i.e. $(S(X), \cdot)$.

$$\text{Let } f, g \in S(X) \quad f \circ g: X \rightarrow X \\ x \mapsto f(g(x)).$$

f, g bijection $\Rightarrow f \circ g$ is a bijection.

$$\Rightarrow f \circ g \in S(X).$$

i.e. $\circ$ is a binary operation on $S(X)$.

$$i) f \circ (g \circ h) = (f \circ g) \circ h \quad \forall f, g, h \in S(X)$$

$$ii) \text{ Defined a map } I: X \rightarrow X, \\ x \mapsto x.$$

$$I \in S(X).$$

$$\& f \circ I = f = I \circ f \quad \forall f \in S(X).$$

$$\Rightarrow I \text{ is the identity of } (S(X), \circ)$$

iii) Let $f \in S(X)$ i.e. f is a bijection from X onto itself.

$$\Rightarrow \exists f^{-1}: X \rightarrow X \text{ defined as}$$

$$f^{-1}(x) = y \quad \text{if} \quad f(y) = x.$$

As f is a bijection, so is f^{-1} . Hence $f^{-1} \in S(X)$.

$$f \circ f^{-1} = I = f^{-1} \circ f \quad \Rightarrow f^{-1} \text{ is the inverse of } f \text{ under } \circ.$$

$$\Rightarrow (S(X), \circ) \text{ is a group.}$$

Is $S(X)$ an abelian group?

Not true in general. Let $a, b, c \in X$

$$\text{Define } f: X \rightarrow X$$

$$a \mapsto b$$

$$b \mapsto c$$

$$c \mapsto a$$

$$x \mapsto x \quad \forall x \neq a, b, c.$$

$$g: X \rightarrow X,$$

$$a \mapsto b$$

$$b \mapsto a$$

$$c \mapsto c$$

$$x \mapsto x \quad \forall x \neq a, b, c$$

So $f, g \in S(X)$.

$$f \circ g: X \longrightarrow X$$

$$a \mapsto c$$

$$b \mapsto b$$

$$c \mapsto a$$

$$x \mapsto x \text{ for } x \neq a, b, c$$

$$g \circ f: X \longrightarrow X$$

$$a \mapsto a$$

$$b \mapsto c$$

$$c \mapsto b$$

$$x \mapsto x \text{ for } x \neq a, b, c$$

$$\text{So } f \circ g \neq g \circ f$$

$\Rightarrow (S(X), \circ)$ is not an abelian group if $|X| \geq 3$.

if $|X| = 2$ then $|S(X)| = 2! = 2$.

I = identity map is one element. & there is one more element say $f \in S(X)$.

$$\text{as } f \circ I = f = I \circ f.$$

$(S(X), \circ)$ is an abelian group.